

Of the estimated 200 billion stars that make up the Milky Way, our home galaxy, only one is vital to our existence: a fairly unremarkable main-sequence star that we call the Sun. As is the case for most other stars in the galaxy, the Sun did not form in isolation. Its mighty gravitational pull keeps hold of a family of bodies that formed with it, the largest of which we know as the planets. Five of these other worlds

AROUND THE **SUN**



have been known since ancient times, from their wanderings among the stars in our night skies. Through the use of telescopes and space-based observatories, two more planets, hundreds of moons, and more than a million smaller objects, including comets and asteroids left over from the Solar System's formation, have also been found. The planets orbit the Sun in a disk, with the paths of smaller bodies generally becoming more scattered farther away from the Sun. The innermost planets—Mercury, Venus, Earth, and Mars—are small, solid globes made up primarily of rock and metals. In contrast, the outer worlds—Jupiter, Saturn, Uranus, and Neptune—are giants formed of gas and liquid, each accompanied by a multitude of their own natural satellites. Despite the extensive knowledge we have of our neighbors, we may not yet fully appreciate the scale of our planetary system. Indeed, large bodies may lie yet undiscovered in the dark extremities of our Solar System.

◀ **Surface of the Sun**

At the high temperatures close to our star, almost all the gas present is split into charged particles—a plasma—through which weaves a tangled jumble of magnetic fields. The complex churning of the Sun's braided surface overlies the primary source of heat and light in our Solar System: the nuclear furnace that is buried in the star's interior.